Department of Biotechnology, School of Energy Technology

B.Tech Biotechnology

Course Matrix for Semester – V (Batch 2024-2028)

Semester	Category Code	Course Code	Course Name	Theory	Tutorial	Practical	Hrs	Credits
Semester 5	OE	MOOC	Open Elective 2 ((NPTEL/SWAYAM/MOOC))	3	0	0	3	3
	HSC	24HS301T	Engineering Economics	3	0	0	3	3
	PE	24BT331T	Medical Diagnostics	3	0	0	3	3
		24BT332T	Green Biotechnology and Pollution Abatement					
		24BT333T	Computer Aided Drug Design					
	PC	24BT301T	Genomics & Proteomics	3	0	2	5	4
	PC	24BT301P	Genomics & Proteomics Laboratory	0	0	2	2	1
	PC	24BT302T	Immunology and Immunotechnology	3	0	2	5	4
	PC	24BT302P	Immunology and Immunotechnology Laboratory	0	0	2	2	1
	PC	24BT303T	Animal and Plant Biotechnology	3	0	2	5	4
	PC	24BT303P	Animal and Plant Biotechnology Laboratory	0	0	2	2	1

24BT331T					Medical Diagnostics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To acquire the knowledge on concepts of medical diagnostics and process involved in the diagnostic laboratories
2. To explore different diagnostic techniques and technologies
3. To understand the molecular diagnostic and imaging tools to assist the clinical diagnosis.
4. To discuss the role of diagnostics in patient management and treatment

UNIT I: FUNDAMENTAL OF DIAGNOSTICS 10 Hrs.

Concepts of diagnosis and its importance in health care, Assessment of clinical history, Laboratory test (Serology, Hematology, Urine & stool), Molecular Diagnostics (PCR), Histological procedures in sample Collection, Preservation & processing of tissue specimens, Fundamental of radiological assessment. Model case presentation.

UNIT II: DIAGNOSTIC MEDICAL MICROBIOLOGY AND RECENT ADVANCES 10 Hrs.

Diagnostic medical microbiology: Microbial pathogenesis, Collection, culturing, identification procedures: Updated immunologic or molecular diagnostic tests. The diagnostic laboratory tests for identification of *Staphylococcus aureus, E. coli, shigella, Salmonella etc.,* including bacteriologic methods for isolation, serologic methods of diagnosis. Test for bacterial Sensitivity tests against antimicrobial agents and its clinical implications and interpretation.

UNIT III: LABORATORY DIAGNOSIS OF VIRUS INFECTION 10 Hrs.

Laboratory Diagnosis of Virus Infection: Viral pathogenesis, Specimen collection and submission, Cultivation & assays for virus, Purification & identification of virus. Immunopathological Methods in Clinical laboratories: Detection of various allergic agents and immunopathology of allergy.

UNIT IV: RADIOLOGY AND ADVANCED IMAGING TECHNOLOGY 10 Hrs.

Radiology and Advanced Imaging Techniques: Introduction to Radiology- X Ray, Ultrasound, MR, CT, PET, Radiographic, High-throughput diagnostics in clinics: DNA chips, diagnosis of genetic disorders, human genome project in diagnostics and Microarrays.

40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Understand the fundamental principles of medical diagnostics
CO2 : Demonstrate various techniques in clinical, histopathological and radiological laboratories
CO3 : Analyze the process to diagnose common infections and interpretation of diagnostic tests and results.
CO4 : Classify and compare the diagnostic techniques and imaging techniques
CO5 : Discuss the role of diagnostics in patient management and treatment.
CO6 : Develop critical thinking skills in evaluating diagnostic approaches and methodologies.

TEXT/REFERENCE BOOKS

1. Elmer W. Koneman, “Color Atlas and Textbook of Diagnostic Microbiology”, Jones & Bartlett Learning.
2. Mary Louise Turgeon, “Linne and Ringsrud’s Clinical Laboratory Science: Concepts, Procedures, and Clinical Applications”, Mosby.
3. Kumar, Abbas, and Aster, “Robbins Basic Pathology”, Elsevier India.
4. Turgeon, Mary Louise. “Clinical Hematology: Theory and Procedures”, Lippincott Williams and Wilkins.
5. Strasinger, Susan King, and Marjorie Schaub Di Lorenzo. “Urinalysis and Body Fluids”,F.A Davis company.
6. Rifai, Nader, Horvath, Andrea R., and Wittwer, Carl T. Tietz , “Textbook of Clinical Chemistry and Molecular Diagnostics”, Elsevier India.
7. Thrall, Donald E.,”Textbook of Veterinary Diagnostic Radiology”, Elsevier.

24BT332T					Green Biotechnology and Pollution Abatement					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To gain the fundamental understanding on environmental degradation
2. To Learn the different aspects of environmental toxicity
3. Understand the monitoring of environmental pollution
4. Understand the environmental issues and their approach on prevention strategies

UNIT I: ENVIRONMENTAL DEGRADATION 10 Hrs.

Environmental degradation, Impact of anthropogenic activities of environment: Land, Water, Air, Forest, Oceans and the coasts; Biodiversity destruction (flora and fauna); Disturbance of ecological balance United Nations Millennium ecosystem assessment. Exponential growth: Four spikes and the reason for spikes, impact and significance of exponential growth, Sustainability of exponential growth, Ecological balance and ecological footprint

UNIT II: ENVIRONMENTAL TOXICOLOGY 10 Hrs.

Toxic chemicals in the environment (air and water) – their effects and biochemical interactions; Biochemical aspects of arsenic, cadmium, lead, mercury, carbon monoxide, ozone and PAN pesticide; Mode of entry of toxic substance, its breakdown and detoxification; biotransformation of xenobiotics; Insecticides / Pesticides in environment, MIC effects. Carcinogens in environment, chemical carcinogenicity, mechanism of carcinogenicity, environmental carcinogenicity testing. Epidemiological issues of toxic compounds and metal poisoning.

UNIT III: ENVIRONMENTAL MONITORING 10 Hrs.

Definition and environmental monitoring process; Sampling – land (site) sampling, water sampling, air sampling, Analysis – physical, chemical and biological analysis methods and process Use of microbial population for environmental monitoring – recombinant DNA technology and proteomics. Monitoring pollution; Bioindicators- markers and genetic indicators; Environment Impact Assessment: EIA complete process, Importance of EIA

UNIT IV: BIOTECHNOLOGICAL METHODS TO POLLUTION ABATEMENT 10 Hrs.

Biofilters, Bioremediation, Biotransformation and Biodegradation: In situ and Ex situ bioremediation; Evaluating Bioremediation; Bioremediation of VOCs. Factors affecting process of biodegradation; Methods in determining biodegradability; Contaminant availability for biodegradation; Use of microbes (bacteria and fungi) and plants in biodegradation and Biotransformation; Phytoremediation: Waste water treatment using aquatic plants; Root zone treatment.

40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Learn the sources of environmental pollution and their causes.
CO2 : Evaluate levels of various pollutants using sophisticated techniques.
CO3 : Understand the advanced treatment methods and pollution abatement techniques
CO4 : Classify the factors and parameters that affect the environmental and economic scenario
CO5 : Develop the assessment of environmental issues pertaining to industry and society
CO6 : Apply the biotechnological based methods for pollution control

TEXT/REFERENCE BOOKS

1. Bruce E Rittman and Perry L McCarty, “Environmental Biotechnology, Principles and Applications”, McGrawhill Higher education, 2020
2. AS Tomar, “Green Biotechnology”, Daya Publishing House, 2019.
3. L K Wang et al. “Environmental Biotechnology”, Vol 10 Handbook of Environmental Engineering, Humana Press, 2010
4. HJ Jördening and J Winter, “Environmental Biotechnology”. WILEY-VCH Verlag Gmbh & Co., 2015

24BT333T					Computer Aided Drug Design					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. Learn the specific information from the enormous and rapidly expanding chemical literature
2. Gain the fundamental on drug/receptor interactions using in silico techniques.
3. Create interest in learning relevant applications in QSAR and Drug Design.
4. Understand the basic SNPs and Computer Aided Drug Design.
5. Learn the concepts of target discovery technology and strategy.

UNIT I: INTRODUCTION 10 Hrs.

Computational biology application, Major databases, Data management, computational molecular biology, Central dogma, Data retrieval, Data mining, Sequence alignment.

UNIT II: PROTEIN STRUCTURE 10 Hrs.

Classification, Visualization, Protein structure database, Visualization tools, Tool for protein ligand interaction, Motif and domains, Protein – protein interaction.

UNIT III: STRUCTURE PREDICTION 10 Hrs.

Methods of sequence based structure prediction, Ab-initio approach for structure prediction, Methods of 2-D structure prediction, Protein function prediction, Homologous modelling. Drug and target discovery technology and strategy, Target validation.

UNIT IV: DRUG DISCOVERY 10 Hrs.

Pharmacogenetics and pharmacogenomics application, SNPs, Important parameter for drug discovery. Drug design approaches, Structure based de novo Methods, QSAR, ADME-Tox property prediction.

40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Classify the computer Aided Drug Design methods and predict drug like compounds.
CO2 : Gain the fundamental practice of Molecular modeling and modern drug discovery.
CO3 : Understand about the rational drug design, based on three-dimensional structures and physicochemical properties of drugs and receptors.
CO4 : List different force fields and their optimization for drug design and simulations.
CO5 : Focus professionally on energy minimization and computer simulation
CO6 : Develop solution to enhance understanding for drug target interactions.

TEXT/REFERENCE BOOKS

1. Rastogi, S. C., "Bioinformatics; Methods and applications; Genomics, Proteomics and Drug Discovery", PHI Learning Publishers, New Delhi, 2021.
2. Gautham N., "Bioinformatics", Narosa Publishing Company, New Delhi, 2020.
3. Vasantha Pattabhi and N.Gautham, "Biophysics", Narosa Publishing Company, New Delhi, 2019.
4. Lesk, A.M., "Introduction to Bioinformatics", Oxford University Press, Oxford, UK, 1st Edition, 2020.
5. Tagelsir Mohamed Gasmelseid, "Pharmacoinformatics and Drug Discovery Technologies: Theories and Applications", Idea Group, 2019. (ISBN: 978-1466603097)

24BT301T					Genomics and Proteomics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	50	50	100

COURSE OBJECTIVES

1. To gain the fundamental knowledge of genome organization
2. To learn the application of genome analysis
3. To understand the concepts of proteomics
4. To learn proteomics analysis and applications

UNIT I: INTRODUCTION TO GENOMICS AND PROTEOMICS 10 Hrs.

Introduction – Organization and structure of genomes, Genome size, Introns and Exons, Sequence complexity, Genome structure in prokaryotes / viruses, Retrofitting. Introduction to Proteomics – The Proteome, Mining proteomes, Bridging Genomics and Proteomics. Proteomics and the new biology.

UNIT II: GENE EXPRESSION AND ANALYSIS 10 Hrs.

Traditional routes of gene identification, detecting open-reading Frames, software programs for finding genes, Identifying the function of a new gene, gene ontology, comparative genomics, Protein structural genomics, determining gene function by sequence comparison and through conserved protein structure Global expression profiling - Analysis of RNA expression, applications of genome analysis and genomics.

UNIT III: PROTEOME ANALYSIS 10 Hrs.

Analysis of proteomes – 2D page and image analysis, Mass spectrometry based methods for protein identification- De novo sequencing using mass spectrometric data, 2-DE gel electrophoresis coupled with mass spectrometry, Micro array techniques, Microarray Technology in Treating Disease.

UNIT IV: APPLICATIONS OF GENOMICS AND PROTEOMICS ANALYSIS 10 Hrs.

Analysis of Genomes of different organisms – Human, Mouse, *Plasmodium falciparum*, *Saccharomyces cerevisiae*, *Mycobacterium tuberculosis*. Application of proteome analysis- drug development and toxicology, Pharmaceutical Applications, Proteomics in drug Discovery in human, phage antibodies as tools, Glycobiology and Proteomics in plant genetics and breeding.

40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Apply the concept of genetic information stored and the current genomic technologies
CO2 : Analyze the significance of mapping with respect to the structure and function of genome
CO3 : Identify the key concepts of proteomics with a mention on various aspects of proteomic studies involving the interactions of these biomolecules with other biomolecules.
CO4 : Choose and compare methods for the purification of proteins and analyzing them on a high through put platform
CO5 : Ponder on the various techniques for expression
CO6 : Harbour the knowledge on some of latest omics concepts

TEXT/REFERENCE BOOKS

1. B. Alberts, R. Herald, A. Johnson, D. Morgan, M. Raff, K. Roberts and P. Walter. “Molecular Biology of the Cell”, 7th Edition, Norton and Co., publishers, Canada, 2022
2. A. Sobti, M. Mukesh and RC Sobti. “Genomics, Proteomics and Biotechnology”, Taylor and Francis, 2022
3. S. B. Primrose and R.M. Twyman – “Principles of Genome Analysis and Genomics”, 8 th Edition, Blackwell Publishing, 2012.

24BT301TP					Genomics and Proteomics Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To view and use the various molecular modelling databases available on the World Wide Web.
2. To retrieve the Nucleic Acids structural modeling
3. To identify the Protein structure function relationship activity
4. To predict Molecular Dynamics simulation

LIST OF EXPERIMENTS

1 Molecular Dynamics simulation
2 Monte Carlo simulation
3 Visualization of Molecular Dynamics
4 Molecular Docking studies
5 Protein structure function relationship activity
6 NGS different Platforms
7 Sequence Data analysis
8 NGS Resources Tools & Databases
9 Whole Genome Sequencing & Exome Sequencing
10 Genome Annotation (gene fusions, isoforms, and structural variants)

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Retrieve a Molecular simulation of tripeptides and carbohydrate
CO2 : Analyse the structural modelling
CO3 : Analyse and retrieve the different visualizations of Molecular Dynamics
CO4 : Analyse different tools and databases of NGS platform
CO5 : Analyse and retrieve different Genome Annotation
CO6 : Explore the algorithms for assembling NGS data

TEXT/REFERENCE BOOKS

1. Megahed Mohammad, “Genomic Data Analysis” LAP Lambert Academic Publishing, 2019
2. Noam Shomrom “Deep Sequencing Analysis”, Springer, 2018
3. JM Bower and H Bolouri, eds, “Computational modeling of genetic and biochemical networks”, MIT Press 2005
4. Orpita Bosu, “Bioinformatics: Experiments, Tools, Databases, and Algorithms” (Oxford Higher Education) by, 2017.
5. Dev Bukhsh Singh (Editor), Rajesh Kumar Pathak “Bioinformatics: Methods and Applications” 1st Edition . 2019
6. Jean-Louis Lassez “Introduction to Bioinformatics” Using Action Labs, 2020

24BT302T					Immunology and Immunotechnology					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	50	50	100

COURSE OBJECTIVES

1. To understand the fundamental concepts of immunology and humoral immunology
2. To know the functions of immune system.
3. Examine the role of cytokines in regulating the immune response
4. Understand the mechanisms underlying autoimmunity and hypersensitivity
5. Investigate the immunological basis of graft rejection

UNIT I: FUNDAMENTAL CONCEPTS OF IMMUNOLOGY 8 Hrs.

History and Scope of Immunology, Innate and Acquired Immunity; Haematopoiesis, Cells of the immune system, Primary and Secondary lymphoid organs, Characteristics of T and B cell receptors, T and B cell maturation, activation and differentiation

UNIT II: HUMORAL IMMUNITY 10 Hrs.

Antigens: Characteristics and Types of Antigens, Factors affecting the immunogenicity, Haptens, Antigenic Determinants, Adjuvants, Vaccines, Routes of Immunization. Theory ofclonal selection, Immunoglobulins: Basic structures, classes and sub classes; Molecular Biology of Immunoglobulin synthesis, Effector functions; Isotope, Allotype, Idiotype, Monoclonal antibodies, Complement system - Alternate, Classical and Lectin pathways

UNIT III: CELL MEDIATED IMMUNITY 14 Hrs.

Structure, types and function of MHC, Phagocytosis, Exogenous and Endogenous pathways of antigen processing and presentation; Cytokines - Structure, function, application and regulation of the immune response. Self-tolerance and possible mechanisms of induction of autoimmunity, Organ specific and systemic autoimmune diseases, IgE mediated (Type I) hypersensitivity, antibody mediated cytotoxic (Type II) hypersensitivity, immune complex mediated (type III) hypersensitivity and delayed type (Type IV) hypersensitivity, Immunological basis of graft rejection

UNIT IV: IMMUNOTECHNOLOGY 8 Hrs.

Antibody affinity and antibody avidity, Crossreactivity, Precipitation reactions, Agglutination reactions - Immunodiffusion and Immunoelectrophoretic techniques, Immunofluorescences, Immunoelectroscopy, RIA, ELISA. Detection of bacterial endotoxins using immunological methods (LAL Test)

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Elaborate the basics of immunity and Immunology
CO2 : Evaluate the role of B cell against extracellular pathogens
CO3 : Illustrate the role of T cell against intracellular pathogens
CO4 : Apply Ag-Ab interactions for diagnostic purposes
CO5 : Identify the conditions due to Immune response against self-antigens
CO6 : Describe the mechanisms involved in different types of hypersensitivity, tolerance and autoimmunity

TEXT / REFERENCE BOOKS

1. Coico, R and Sunshine, G. (2009)," Immunology: A Short Course" (6th ed.), John Wiley& sons, Inc (New Jersey), ISBN: 978-0-470-08158-7.
2. Jenni Punt, Sharon Stranford, Patricia Jones, Judith A Owen," Kuby J, Immunology",; Edition, 8 ; Publisher, Macmillan Learning, 2018
3. Janeway's Immunobiology - 9th Edition. 2017, Garland Science
4. Peter J. Delves, Seamus J. Martin, Dennis R. Burton, and Ivan M. Roitt, Roitt's Essential Immunology"2017, Wiley-Blackwell

24BT302P					Immunology and Immunotechnology Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

1. Understand the principles and techniques involved in isolating lymphocytes from blood or spleen.
2. Gain practical experience in performing agglutination reactions
3. Familiarize with enzyme-linked immunosorbent assay (ELISA) and DOT ELISA techniques for antigen-antibody detection
4. Learn how to determine antibody titers through experimental procedures.

LIST OF EXPERIMENTS

1 Isolation of lymphocytes from blood / spleen.
2 Purification of immunoglobulins from serum
3 Assays based on precipitation reactions - Ouchterlony double immunodiffusion (DID) and Mancini radial immunodiffusion (SRID).
4 Assays based on agglutination reactions - Blood typing (active) & passive agglutination
5 Enzyme linked immunosorbent assay (ELISA) & DOT ELISA)
6 Precipitation assays:Immunodiffusion /immunoelectrophoresis
7 Antibody Titer
8 Western blotting
9 Preparation of antigen and adjuvant emulsion (mineral oil, alum, Freund's adjuvants,etc.)
10 Immunization of mouse through intra muscular and intra peritoneal routes

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Relate various immunological components in body's defence mechanism
CO2 : Demonstrate functions of cells and organs of the immune system
CO3 : Understanding of immune responses and methods of clinical diagnosis for identifying Ag-Ab interactions
CO4 : Interpret Western blotting results to identify specific proteins
CO5 : Assess the quality of antigen-adjuvant emulsions prepared for immunization
CO6 : Design experimental protocols for isolation of lymphocytes

TEXT / REFERENCE BOOKS

1. Coico, R and Sunshine, G. (2009)," Immunology: A Short Course" (6th ed.), John Wiley& sons, Inc (New Jersey), ISBN: 978-0-470-08158-7.
2. Jenni Punt, Sharon Stranford, Patricia Jones, Judith A Owen," Kuby J, Immunology",; Edition, 8 ; Publisher, Macmillan Learning, 2018
3. Janeway's Immunobiology - 9th Edition. 2017, Garland Science
4. Peter J. Delves, Seamus J. Martin, Dennis R. Burton, and Ivan M. Roitt, Roitt's Essential Immunology"2017, Wiley-Blackwell

24BT303T					Animal and Plant Biotechnology					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	50	50	100

COURSE OBJECTIVES

1. To gain the fundamental knowledge of biotechnology
2. To create awareness in Plant and Animal biotechnology.
3. To impart knowledge in micromanipulation techniques in cell culture.
4. To understand and apply the principles of transgenic plants and animals.

UNIT I: INTRODUCTION TO PLANT CELL CULTURE 10 Hrs.

Introduction – culturing plant and animal cells -Tissue Culture media, Callus and suspension culture, Somoclonal Variation, Micro propagation, Organogenesis, Somatic embryogenesis, transfer and establishment of whole plants in soil, green house technology.

UNIT II: INTRODUCTION TO ANIMAL CELL CULTURE 10 Hrs.

Chemically defined and serum free media. Laboratory design, Primary cell culture, Establishment of cell line, Maintenance and Preservation of cell line. Characterization, Cross contamination, Scale up of Cell cultures for Product development

UNIT III: DESIGN OF GENE CONSTRUCT AND ADVANCED TECHNOLOGIES 10 Hrs.

Plant selectable markers; Reporter genes; Positive selection; Transgene silencing; Strategies to avoid transgene silencing; advanced technologies – cis genesis and intragenesis; RNAi technology, genome editing technology, CRISPR/Cas.

UNIT IV: TRANSGENICS PLANTS AND ANIMALS – APPLICATIONS 10 Hrs.

Plant Genetic Transformation Methods: Chloroplast transformation and its advantages. Application of Plant Genetic transformation: Herbicide resistance: Insect resistance, Disease resistance antifungal proteins. Embryo transfer-Micromanipulation technology, In Vitro fertilization, Transgenic Animals and their significance. Ethical issues in transgenic plants and animals

40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Demonstrate the propagation of plant cultures
CO2 : Perform the maintenance and culturing of animal cell
CO3 : Understand the principles of gene construct
CO4 : Get acquainted with the ethical issues involved in transgenic organisms
CO5 : Focus professionally on latest technologies like genome editing
CO6 : Apply their knowledge on transgenic plants in day to day life

TEXT/REFERENCE BOOKS

4. R. Ian Freshney, “Culture of Animal Cells: A Manual of Basic Technique and Specialized Applications”, Eight Edition. Publisher, John Wiley & Sons, 2021.
5. Animal cell culture 3rd ed., by John R.W. “Masters A Practical Approach” Wiley Blackwell Publishers – 2010
6. S. Park. “Plant Tissue Culture: Techniques and Experiments”, Fourth Edition, Elsevier science publishers, 2021.
7. Chawala. H.S., “Introduction to plant Biotechnology”, Third Edition, Oxford and IBH Publishing Co. Pvt. LTD.New Delhi 2021.
8. Grierson,D. “Plant Biotechnology in Agriculture Prospects for the 21st Century”, Academic press, 2012

24BT303P					Animal and Plant Biotechnology Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To introduce students to the principles of animal and plant cell culture, emphasizing biosafety and ethical practices.
2. To provide hands-on experience in preparing media, maintaining cell cultures, and cryopreserving cell lines.
3. To train students in detecting contaminants and evaluating cell viability and cytotoxicity using standard assays.
4. To teach advanced molecular techniques such as transfection, transformation, and DNA fingerprinting.
5. To develop skills in plant tissue culture, including creating transgenic plants and regenerating tissues for research and applications.

LIST OF EXPERIMENTS

1 Animal Cell Culture Biosafety and Ethical Concerns
2 Preparation of reagents and media for cell culture
3 Cell counting and plating
4 Preparation of primary cultures
5 Performance of cell passaging
6 Maintenance of Adherent (Monolayer) and Suspension Cell culture
7 Cryopreservation of cell lines
8 Determination of common cell culture contaminants
9 Cell Viability Assay (MTT reagent)
10 Preparation and sterilization of standard tissue culture media.
11 Sterilization of explants and generation of undifferentiated mass of cells.
12 Regeneration of plants from Callus Culture.
13 Agrobacterium mediated transformation of plants.
14 Selection and screening of transgenic plants.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Gain knowledge on Biosafety and Ethical Concerns
CO2 : Perform primary culture and subculture
CO3 : Preparation and sterilization of standard tissue culture media
CO4 : Understand the importance of sterilization in cell culture
CO5 : Gain knowledge on Cell Viability and toxicity
CO6 : Develop skills on basic cell culturing techniques

TEXT/REFERENCE BOOKS

1. R. Ian Freshney. “Culture of Animal Cells: A Manual of Basic Technique and Specialized Applications”, Eight Edition. Publisher, John Wiley & Sons, 2021
2. George Acquaah. “Principles of Plant Genetics and Breeding”. Third Edition, Blackwell Publishing, 2020